

UDAY SONAWANE

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Master of Science, Computer Science | **GPA 4.0/4.0**

Sept 2024 to May 2026

Coursework: Programming Design Paradigm, Algorithms, DBMS, Cloud Computing, Deep Learning, Foundations for Generative AI, Foundations of AI, Machine Learning

Graduate Teaching Assistant, CS 5200 Database Management Systems (3 semesters; mentor 60+ students per cohort, author rubrics, onboard incoming TAs)

Vellore Institute of Technology

Vellore, India

Bachelor of Technology, Computer Science | **GPA 3.65/4.0**

Aug 2019 to Apr 2023

Coursework: Software Engineering, Mathematical Model for Data Science, Operating Systems, Parallel and Distributed Computing

EXPERIENCE

Siemens Digital Industries Software

Pune, India

Software Engineer Intern

June 2023 to Jan 2024

- Owned end-to-end debugging of production C++ and React services across distributed components, triaging system and console log failures and shipping fixes through CI/CD without breaking dependent teams' code paths.
- Acted as point of contact between the platform team and 3 adjacent product teams: scoped trade-offs on cross-cutting bugs, aligned on resolution strategy, and translated agreements into shipped patches.
- Built a Python ETL pipeline to extract, normalize, and classify large-scale system logs, cutting manual debugging time by approximately 60% and surfacing recurring failure modes.
- Built automated test scaffolding to expand regression coverage on defect-prone modules and maintained SDLC traceability in Polarion for safety-critical components.

PROJECTS

FullStack-StockVision, Stock Forecasting Platform | Python, C++, ONNX Runtime, React, FastAPI | [GitHub](#)

- Built a stock-forecasting platform for any yfinance ticker, with a Python training CLI, a FastAPI prediction service, and a parallel C++ ONNX-Runtime server exposing the same REST API; React frontend consumes either backend interchangeably.
- Shipped 7 model architectures (LSTM, BiLSTM, GRU, CNN-LSTM, Transformer, TCN, linear baseline) behind a plugin-style registry so a new architecture is a one-file change, with a pytest suite for Python and CTest for the C++ server.

ArchPlexBench, From-Scratch Sequence-Model Benchmark | PyTorch, Mamba, Transformers, LSTM, CUDA | [GitHub](#)

- Implemented 5 sequence architectures from scratch in pure PyTorch (decoder-only and encoder-decoder Transformers, LSTM, Mamba, multi-head Mamba) at matched 6M parameters and benchmarked them on WikiText-2.
- Built a torch.compile-fused Hillis-Steele parallel prefix scan to make the Mamba SSM tractable without the upstream CUDA kernel; Mamba reached test perplexity 215, around 3 times lower than the Transformer baseline at matched compute.

SimpleRayTracer, C++ Rendering Engine | C++, CMake | [GitHub](#)

- Built a recursive ray-tracing engine from scratch covering vector math, ray-sphere intersection, materials, and anti-aliasing, with a modular CMake structure under SOLID so new materials and primitives plug in without touching the renderer core.

Generative Engine Optimization, Multi-LLM Research Project | Python, ChromaDB, OpenRouter, RAG | [GitHub](#)

- Co-developed a research project studying brand visibility across 4 commercial LLMs over 240 queries and 3 conditions; designed the experimental framework and proposed the pseudo-brand methodology.
- Built the RAG pipeline using sentence-transformers, ChromaDB, and category-filtered retrieval; a fictional brand reached 95% mention rate at position 1 across LLMs through optimized RAG content alone.

SKILLS

Languages: C++, Python, Java, JavaScript, SQL

Systems and Backend: REST APIs, Multithreading, Memory Management, Profiling, Software Design Patterns, OOP, Concurrency

Web and Cloud: React, Flask, Django, Node.js, AWS (EC2, S3), Docker, Git, CI/CD, ETL

Foundations: Data Structures and Algorithms, Operating Systems, Parallel and Distributed Computing, DBMS, SDLC

LLM and GenAI: RAG, LangGraph, LangChain, ChromaDB, sentence-transformers, prompt caching, fine-tuning, Anthropic API

ACHIEVEMENTS

Codeforces 1439 (top 502 of 23,000 in Round 944); Google Data Analytics, UC San Diego Algorithmic Toolbox, HackerRank Problem Solving (Intermediate)